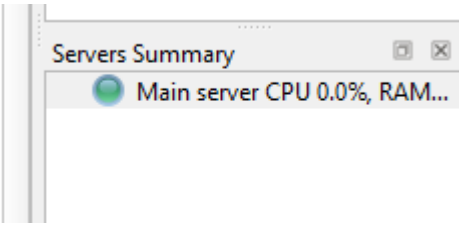
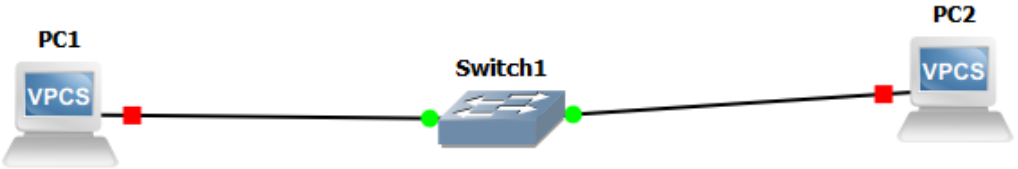
1) Установить и настроить эмулятор GNS3C

Сделано и подключено к серверу



2) Создать простейшую сеть, состоящую из 1 коммутатора и 2 компьютеров, назначить им произвольные ip адреса из одной сети



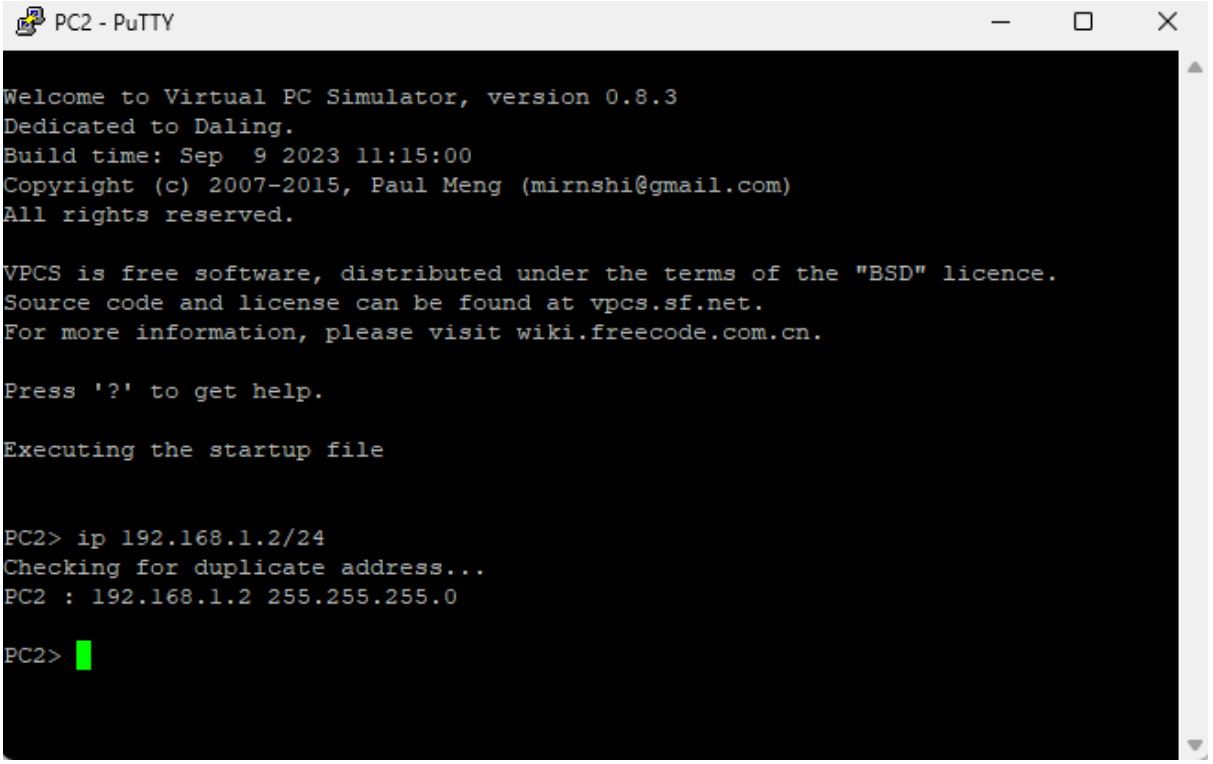

```
PC1> ip 192.168.1.1/24
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1>
```

PC1> ip 192.168.1.1/24

Checking for duplicate address...

PC1 : 192.168.1.1 255.255.255.0



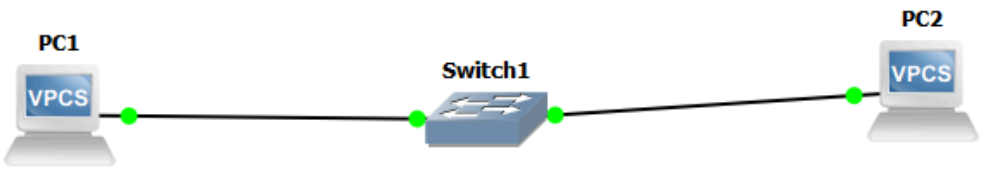
PC2> ip 192.168.1.2/24

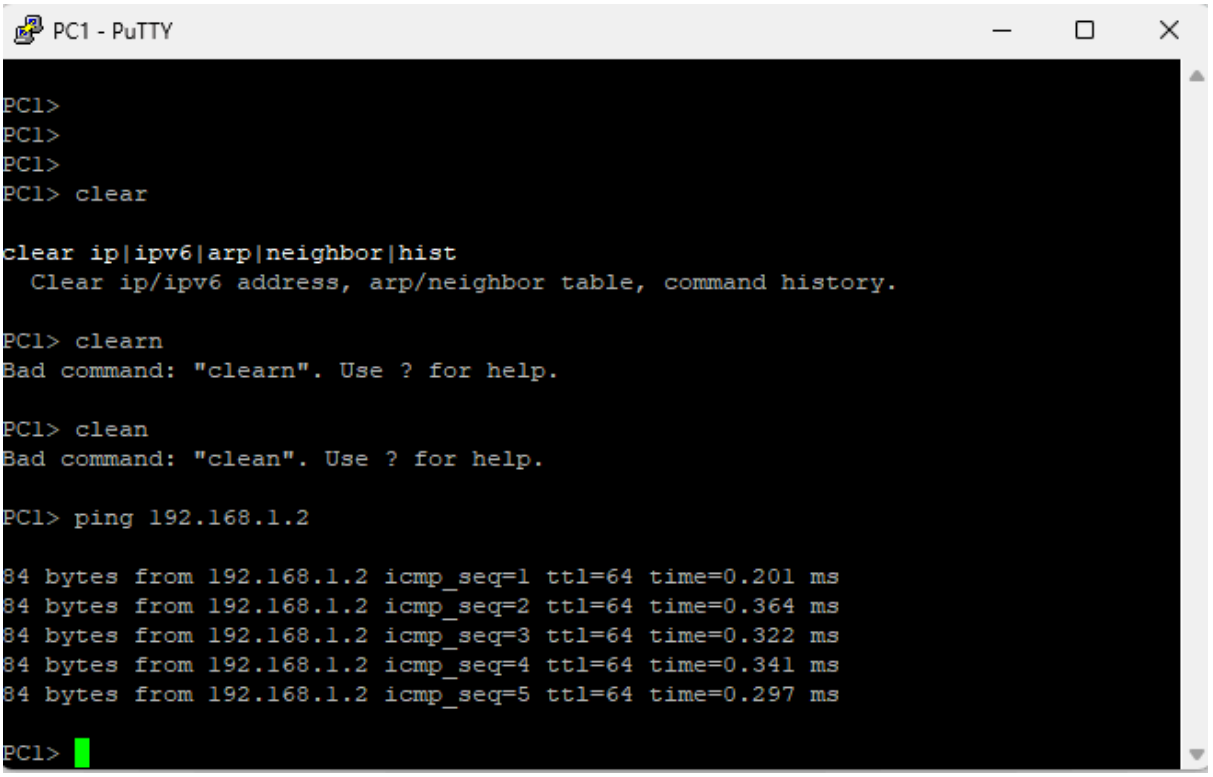
Checking for duplicate address...

PC2 : 192.168.1.2 255.255.255.0

PC2>

3) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера





PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=0.201 ms

84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=0.364 ms

84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.322 ms

84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.341 ms

84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=0.297 ms

4) Перехватить трафик протокола arp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

Со стороны PC1-Switch(request)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.2? Tell 192.168.1.1

Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

.... ..1. .... = LG bit: Locally administered address (this is NOT the factory default)

....1 .... = IG bit: Group address (multicast/broadcast)

Source: Private_66:68:00 (00:50:79:66:68:00)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 0]

Padding: 000000000000000000000000000000000000

Frame check sequence: 0x00000000 [unverified]

[FCS Status: Unverified]

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)

Sender IP address: 192.168.1.1

Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)

Target IP address: 192.168.1.2

Со стороны PC1-Switch(reply)

2	0.000149	Private_66:68:01	Private_66:68:00	ARP	64 192.168.1.2 is at 00:50:79:66:68:01

Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)

Destination: Private_66:68:00 (00:50:79:66:68:00)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Source: Private_66:68:01 (00:50:79:66:68:01)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 1]

Padding: 000000000000000000000000000000000000

Frame check sequence: 0x00000000 [unverified]

[FCS Status: Unverified]

Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)

Sender IP address: 192.168.1.2

Target MAC address: Private_66:68:00 (00:50:79:66:68:00)

Target IP address: 192.168.1.1

Со стороны PC2-Switch(request)

No.	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.2? Tell 192.168.1.1

Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

.... ..1. .... = LG bit: Locally administered address (this is NOT the factory default)

....1 .... = IG bit: Group address (multicast/broadcast)

Source: Private_66:68:00 (00:50:79:66:68:00)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 0]

Padding: 000000000000000000000000000000000000

Frame check sequence: 0x00000000 [unverified]

[FCS Status: Unverified]

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)

Sender IP address: 192.168.1.1

Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)

Target IP address: 192.168.1.2

Со стороны PC2-Switch(reply)

2	0.000177	Private_66:68:01	Private_66:68:00	ARP	64 192.168.1.2 is at 00:50:79:66:68:01

Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)

Destination: Private_66:68:00 (00:50:79:66:68:00)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Source: Private_66:68:01 (00:50:79:66:68:01)

.... ..0. .... = LG bit: Globally unique address (factory default)

....0 .... = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 1]

Padding: 000000000000000000000000000000000000

Frame check sequence: 0x00000000 [unverified]

[FCS Status: Unverified]

Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)

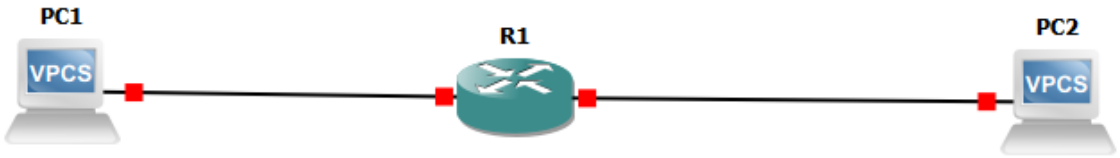
Sender IP address: 192.168.1.2

Target MAC address: Private_66:68:00 (00:50:79:66:68:00)

Target IP address: 192.168.1.1

Анализ: сначала отправляется широковещательный запрос дабы узнать где находится конкретный мак-адрес, потом тот кто имеет нужный мак-адрес отзывается и отправляет напрямую тому, кто вопрошал.

5) Создать простейшую сеть, состоящую из 1 маршрутизатора и 2 компьютеров, назначить им произвольные ip адреса из разных сетей




```
PC1> ip 192.168.1.1/24 192.168.1.254
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0 gateway 192.168.1.254

PC1>
```

PC1> ip 192.168.1.1/24 192.168.1.254

Checking for duplicate address...

PC1 : 192.168.1.1 255.255.255.0 gateway 192.168.1.254

PC1>

```
PC2> ip 192.168.2.2/24 192.168.2.254
Checking for duplicate address...
PC2 : 192.168.2.2 255.255.255.0 gateway 192.168.2.254

PC2>
```

PC2> ip 192.168.2.2/24 192.168.2.254

Checking for duplicate address...

PC2 : 192.168.2.2 255.255.255.0 gateway 192.168.2.254

PC2>

```
R1#enable
R1#config
R1#configure ter
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#inter
R1(config)#interface fast
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip ad
R1(config-if)#ip address 192.168.1.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
*Mar  1 00:01:50.815: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar  1 00:01:51.815: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config-if)#^Z
```

R1#enable

R1#config

R1#configure ter

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#inter

R1(config)#interface fast

R1(config)#interface fastEthernet 0/0

R1(config-if)#ip ad

R1(config-if)#ip address 192.168.1.254 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#

*Mar 1 00:01:50.815: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:01:51.815: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

```
R1(config)#interface fastEthernet1/0
R1(config-if)#ip address 192.168.2.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#'
*Mar  1 00:07:49.911: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Mar  1 00:07:50.911: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
```

R1(config)#interface fastEthernet1/0

R1(config-if)#ip address 192.168.2.254 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#'

*Mar 1 00:07:49.911: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up

*Mar 1 00:07:50.911: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

6) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера

```
PC1> ping 192.168.2.2

192.168.2.2 icmp_seq=1 timeout
84 bytes from 192.168.2.2 icmp_seq=2 ttl=63 time=12.846 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=63 time=15.533 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=63 time=15.603 ms
84 bytes from 192.168.2.2 icmp_seq=5 ttl=63 time=15.865 ms
```

PC1> ping 192.168.2.2

192.168.2.2 icmp_seq=1 timeout

84 bytes from 192.168.2.2 icmp_seq=2 ttl=63 time=12.846 ms

84 bytes from 192.168.2.2 icmp_seq=3 ttl=63 time=15.533 ms

84 bytes from 192.168.2.2 icmp_seq=4 ttl=63 time=15.603 ms

84 bytes from 192.168.2.2 icmp_seq=5 ttl=63 time=15.865 ms

7) Перехватить трафик протокола arp и icmp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

PC1 - Router

No.	Time	Source	Destination	Protocol	Length	Info
5	21.241274	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.254? Tell 192.168.1.1
6	21.243578	cc:01:10:98:00:00	Private_66:68:00	ARP	60	192.168.1.254 is at cc:01:10:98:00:00

request

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)

Sender IP address: 192.168.1.1

Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)

Target IP address: 192.168.1.254

reply

Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: cc:01:10:98:00:00 (cc:01:10:98:00:00)

Sender IP address: 192.168.1.254

Target MAC address: Private_66:68:00 (00:50:79:66:68:00)

Target IP address: 192.168.1.1

No.	Time	Source	Destination	Protocol	Length	Info
7	21.244458	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x464f, seq=1/256, ttl=64 (reply in 8)
8	21.273810	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x464f, seq=1/256, ttl=63 (request in 7)
9	22.274553	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x474f, seq=2/512, ttl=64 (reply in 10)
10	22.290576	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x474f, seq=2/512, ttl=63 (request in 9)
11	23.291763	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x484f, seq=3/768, ttl=64 (reply in 12)
12	23.307140	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x484f, seq=3/768, ttl=63 (request in 11)
14	24.307887	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x4a4f, seq=4/1024, ttl=64 (reply in 15)
15	24.323958	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x4a4f, seq=4/1024, ttl=63 (request in 14)
16	25.325101	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x4b4f, seq=5/1280, ttl=64 (reply in 17)
17	25.340675	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x4b4f, seq=5/1280, ttl=63 (request in 16)

request

Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0xd9bb [correct]

[Checksum Status: Good]

Identifier (BE): 17999 (0x464f)

Identifier (LE): 20294 (0x4f46)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Response frame: 8]

Data (56 bytes)

reply

Internet Control Message Protocol

Type: Echo (ping) reply (0)

Code: 0

Checksum: 0xe1bb [correct]

[Checksum Status: Good]

Identifier (BE): 17999 (0x464f)

Identifier (LE): 20294 (0x4f46)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Request frame: 7]

[Response time: 29.352 ms]

Data (56 bytes)

Router - PC2

No.	Time	Source	Destination	Protocol	Lengtl	Info
5	18.294810	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.2.254? Tell 192.168.2.2
6	18.304671	cc:01:10:98:00:10	Private_66:68:01	ARP	60	192.168.2.254 is at cc:01:10:98:00:10

request

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)

Sender IP address: 192.168.2.2

Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)

Target IP address: 192.168.2.254

reply

Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: cc:01:10:98:00:10 (cc:01:10:98:00:10)

Sender IP address: 192.168.2.254

Target MAC address: Private_66:68:01 (00:50:79:66:68:01)

Target IP address: 192.168.2.2

No.	Time	Source	Destination	Protocol	Lengtl	Info
4	18.294576	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x464f, seq=1/256, ttl=63 (reply in 7)
7	18.305459	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x464f, seq=1/256, ttl=64 (request in 4)
8	19.321420	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x474f, seq=2/512, ttl=63 (reply in 9)
9	19.321644	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x474f, seq=2/512, ttl=64 (request in 8)
10	20.337984	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x484f, seq=3/768, ttl=63 (reply in 11)
11	20.338254	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x484f, seq=3/768, ttl=64 (request in 10)
13	21.354819	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x4a4f, seq=4/1024, ttl=63 (reply in 14)
14	21.355074	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x4a4f, seq=4/1024, ttl=64 (request in 13)
15	22.371533	192.168.1.1	192.168.2.2	ICMP	98	Echo (ping) request id=0x4b4f, seq=5/1280, ttl=63 (reply in 16)
16	22.371780	192.168.2.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0x4b4f, seq=5/1280, ttl=64 (request in 15)

request

Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0xd9bb [correct]

[Checksum Status: Good]

Identifier (BE): 17999 (0x464f)

Identifier (LE): 20294 (0x4f46)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Response frame: 7]

reply

Internet Control Message Protocol

Type: Echo (ping) reply (0)

Code: 0

Checksum: 0xe1bb [correct]

[Checksum Status: Good]

Identifier (BE): 17999 (0x464f)

Identifier (LE): 20294 (0x4f46)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Request frame: 4]

[Response time: 10.883 ms]

Data (56 bytes)

Анализ: пакеты успешно прошли по протоколу icmp с двух сторон, идентификаторы совпадают, контрольные суммы тоже, поменялись только arp запросы, они идут через маршрутизатор и запрашивают его ip